Security and Privacy GroupX

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1 Anonymization

Each of the features present in the original dataset is anonymized in the context of the analysis results discussed in section 3: after a first, stricter anonymization, the methods, parameters, and overall choices are reviewed to ensure a fair level of utility is retained, at the cost of higher risk.
A per-column description of the final anonymization setup follows, starting from the quasi-identifiers, and ending with the sensitive variable `party` (the person's political preference).
It is also worth mentioning that, in many cases, the description of properties and distributions of pre-anonymization features will be kept quite vague to avoid leaking information about the uncensored data.
All columns that are shared between the public population record and the private survey data are considered quasi-identifiers, as their knowledge is public and they can be used to re-identify survey participants.

1.1 Date of Birth

The `dob` column contains the date of birth of each person.
The precision of such information is quite an important threat to privacy: it is likely that very few people share a birth date, resulting in a substantial amount of sample and population uniques (especially when combined with other quasi-identifiers), contributing to poor risk metrics.
To avoid this, the `dob` column is subject to two layers of generalization: from date of birth to age, and from age to age group.
Each layer pools many values from the previous layer together to ensure a decrease in the number of unique entries; many different DOBs result in the same age, and many ages result in the same age group, as exemplified in Table 1.

While the conversion from DOB to age is an objective computation, determining the amount and size of the age group is arbitrary.
By following age quantiles, we propose a 5-group segmentation of the data, which ensures balanced group size while still presenting itself as a logical binning for ages;

Date of Birth	Age	Age Group
15 August 1984		
10 December 1984	40	
05 May 1985		35–44
20 September 1986		
14 February 1987	38	

Table 1: An example of how generalization reduces the number of unique values. Age is computed at the moment of data collection (01/07/2025).

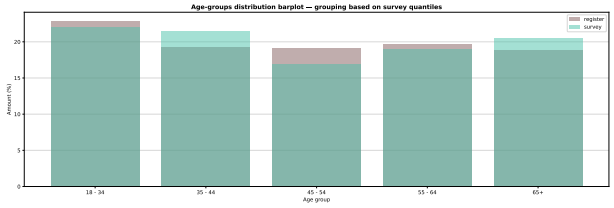


Figure 1: Barplot distribution for age-group values for both survey sample and register population.

the groups are `18-34`, `35-44`, `45-54`, `55-64`, and `65+`. As we can see, the extreme groups contain younger and older people, respectively, while intermediate groups all span 10 years.

To conclude, the generalization steps just described reduce the almost 200 unique dates of birth to 5 age groups, massively reducing risk without an excessive loss of utility (as will be explored in section 3).

1.2 Sex

The column `sex` is a binary variable, with levels `Male` and `Female`.

PRAM is applied with $p = 0.10$: each record has a 10% chance of flipping its `sex` value. This guarantees a modest utility retention by avoiding excessive data manipulation, while still ensuring that potential attackers can not rely safely on the `sex` feature to link records, as there is a chance it contains wrong information.

Overall, about 20 entries change their recorded value, without altering the total count of males and females significantly.

	Published as:	
Original	**Male**	**Female**
Male	90%	10%
Female	10%	90%

Table 2: PRAM Transition Probability Matrix for the `sex` column.

1.3 e-Vote

The `evote` column is a binary variable, with levels `0` (corresponding to votes cast on paper), and `1` (corresponding to votes cast electronically).

The `evote` feature is anonymized via PRAM, just as described with `sex` in subsection 1.2. Instead of a 10% chance of flipping, each entry has a 15% probability, resulting in an even more unreliable column.

This increase in privacy is possible because, as we will see in section 3, the QI `evote` is

not significant in any of the analyses carried out, and can therefore be manipulated more without the danger of losing much utility.
About the expected number of entries are modified, with a final value distribution (114 on-paper votes, 86 electronic votes) that does deviate from the original more than what observed with `sex`.

	Published as:	
Original	**0**	**1**
0	85%	15%
1	15%	85%

Table 3: PRAM Transition Probability Matrix for the `evote` column.

1.4 Marital Status

Relatively little changes are applied to the `marital_status` column.
This categorical feature originally presents 4 levels: `Married/separated`, `Never married`, `Divorced`, and `Widowed`. The last two are particularly uncommon in the survey data and are therefore grouped, resulting in a new level `Widowed / Divorced` that, once again, reduces the risk associated with the rarity of the original values.
While a coarser grouping was considered (namely a `Married/separated`, `Not married` binary grouping), in order to retain enough utility most of the column effectively remains untouched.

1.5 Citizenship

As some might intuitively expect, the overwhelming majority of raw `citizenship` entries are `Danish`, while other nationalities are often present only once, and at most a few times. Many sample uniques in the raw data can therefore be imputed on this feature alone.
Once more, the chosen anonymization strategy for this categorical feature is grouping, resulting in a binary `citizenship` column with two possible values: `Danish` and `Non-Danish`, with 177 and 23 occurrences, respectively.

1.6 Party Voted

As the main sensitive variable, `party` contains information about what political faction the person voted for. The possible values are `Green`, `Red`, and `Invalid vote`
While the main goal of anonymization is to avoid record linkage through quasi-identifiers, i.e. attackers able to pair an entry in the survey data with a person's public register data, we can apply its principles on `party` as a last line of defense: even in the scenario where a record is de-anonymized, the recovered political preference is not guaranteed to be accurate.
Practically, PRAM is applied once again, but with a slight difference from the setup used for `sex` and `evote`. Invalid votes have an equal chance, 4%, of becoming either Green or Red votes. Every other record has a 7% chance of changing its vote (from Green to

Red or vice versa), and a 1% chance of becoming `Invalid vote`.
The altered distribution closely resembles the original distribution, and a relatively large number of entries see their `party` value change.

	Published as:		
Original Vote	**Green**	**Red**	**Invalid**
Green	92%	7%	1%
Red	7%	92%	1%
Invalid	4%	4%	92%

Table 4: PRAM Transition Probability Matrix for the `party` column.

1.7 Education

While technically also a sensitive variable, as it is not public data present in the population register, the `education` column is left unmodified to retain utility.
Various groupings have been tried, but all determined an excessive shift in analysis results and were therefore discarded.

1.8 Dropped Columns

Both the `name` and `zip` columns are completely discarded from the data.

The column `name` is a direct identifier. We do not replace it with a pseudonym, since the register contains the names of the whole population, an attacker could encrypt or hash every name in it in the same way and link the records again.

Moreover `zip` is also a quasi-identifier, and in the register it matches with the polling station of each person. It only has 4 values and combined with the other attributes, it splits the survey into many more and much smaller groups. If we keep it as is it would raise the mean identification risk from 0.095 to about 0.27, the amount of sample uniques from 13% to about 45% and the share of population uniques from 1% to about 11%. We also tested PRAM on `zip` (flip probability 0.15). It did not lower these results. Generalizing the column wouldn't be helpful either since with only 4 values any coarser groups will almost nullify utility, giving a similar result as if we remove it. We therefore chose to suppress the variable.

Clearly potential analysts won't be able to split the survey by polling station, consequently losing some possible information. This is the cost of the suppression.

2 Risk metrics

We presume an adversary who has acccess to the anonymized file, the register which is public, the election results and this report.

The quasi-identifiers are the attributes that the survey and the register have in common, which are: sex, age group, citizenship, marital status and voting channel.

All the risk metrics presented are computed using all the quasi-identifiers.

2.1 Sample Frequency

For each row, the sample frequency f is equal to the total amount of rows in the survey dataset that share the same QI values: a higher f determines a more commonly occurring record 'archetype'.
A high average f can be a sign of a secure dataset. A person's QI values (obtained from the public registry) can be matched to any of the f survey rows that present those same values: with a higher f, it is very hard to determine which of the entries is the one actually linked to the person.
k-anonymity is a related concept, and simply expresses the minimum sample frequency of the dataset.
The anonymized survey only adheres to 1-anonymity, which can seem worrying at first. In fact, 13% of the samples have $f = 1$ (and are therefore sample uniques).
More reassuring figures are the average f, equal to 4.93, the share of entries with $f > 5$, equal to 39.5%, and the share of entries with $f > 9$, equal to 10.5%.
Overall, while it is true that a part of the dataset has a concerningly low sample frequency, an important portion of the data sits at a higher, safer f value (up to a maximum of 11).
For a summary of sample frequency related metrics, consult Table 5, and see Figure 2 for the complete sample frequency distribution.

Min.	**1**
Average	**4.93**
Max.	**11**
Entries with $f > 5$	**39.5%**
Entries with $f > 9$	**10.5%**

Table 5: Salient sample frequency statistics.

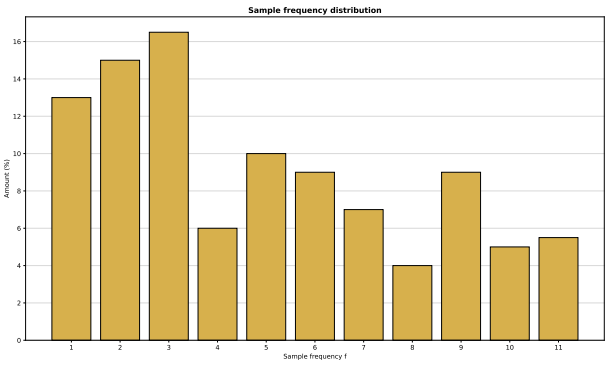


Figure 2: Barplot distribution for sample frequency values.

2.2 Re-identification Risk

To properly present the re-identification risk, a small parenthesis about population frequency F is needed. Population frequency is very similar to sample frequency, with the only difference that, for a record in our survey data, F is equal to the amount of records *in the population register* that present the same combination of QI values. The principles are the same, a higher F is desirable for privacy. Re-identification risk r is derived from population frequency via the simple formula $r = 1/F$. A lower risk for a given survey entry signifies a high count of individuals in the register that share its same QI variables, complicating the task of discovering which of the candidate people is 'hidden' behind that entry.

The highest risk in the anonymized survey data is 1: this would normally mean that such entries can be re-identified immediately (they match exactly one register entry), but due to the use of perturbative methods such as PRAM (described in subsection 1.2, subsection 1.3), the attackers are left with some degree of uncertainty, and cannot guarantee that the record linkage is correct. In addition, the perturbation of the sensitive variable `party` (subsection 1.6) prevents the attackers from confidently uncovering the person's political affiliation for the same reasons, even in case of a successful de-anonymization of a record.

The average risk is decently low, equal to 0.095, with 10% of entries above $r = 0.2$, and only 1% above $r = 0.5$ (all entries above this last threshold are the population uniques, we do not register risk values in the range $(0.5, 1)$).

For a comprehensive view of risk values in the anonymized survey data, refer to Figure 3.

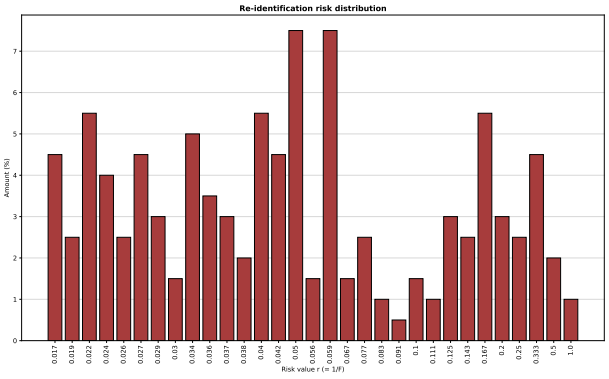


Figure 3: Barplot distribution for risk values.

3 Analysis

The statistical analysis conducted on the data makes use of different tests, described below. A comprehensive review of analysis results then follows.

It is important to note from the beginning how the anonymization process presented in section 1 exclusively results in attenuated correlations: no case of spurious statistical significance emerges from the way random chance shapes the raw data; therefore, we call for analysts using our published data to consider using slightly less harsh p-value threshold for significance, to account for the weakened relationship between features.

3.1 Methods for the continuous variable Age

In the raw data the column age can be used without grouping it into intervals, thus it becomes a reference to understand the information loss we have when we use the five bands in the anonymized file.

For each one of the outcomes in the columns subject to analysis (Green vs Red, E-vote vs Polling Station Vote), we fit a logistic regression using age as the only feature. The values of the age are integers computed from the date of the survey and then divided by 10, thus the odds ratio can be read decade by decade.

3.2 Methods for categorical variables

In order to accurately investigate the relationship between voting behaviors and demographic characteristics, we use some standard statistical tests. A key test is the Pearson's Chi Square test of Independence. The test is utilised to check whether two categorical variables have a significant relationship in a population. This is required to successfully investigate whether political preferences (Part B) and voting channel choices (Part C) are associated with any of the given demographic attributes such as sex, education, marital status, ZIP code, and citizenship. In addition, we calculate Cramer's V to quantify the strength of these associations regardless of their statistical significance. Some of the categories contain a relatively small number of observations, in particular, invalid votes and rare citizenships. In these cases, we use permutation tests (5,000 iterations) to get more reliable p-values. Statistical significance is tested at the 5 percent level, after p-values are adjusted for multiple testing.

3.3 Comparison of political preferences in survey data and election results

A chi-squared test was used to compare the survey votes with the election results for each voting location, with the p-value as the indicator of a significant difference. The resulting p-values varied considerably across the voter locations, showing high p-values for sections: 2200 ($p = 0.2314$), 2400 ($p = 0.619$) and e-votes ($p = 0.4896$), meaning no statistically significant differences were detected between the survey and the election results there. Low p-values were obtained for sections: 2100 ($p = 0.0002$) and 2300 ($p = 0.0000$), indicating a statistically significant difference between the survey and the election results in those locations.

The same analysis on the non-anonymized survey data produces identical p-values on every section, therefore the anonymization process did not affect the results of this analysis.

We also analyzed the survey per location against the elections per location. This yielded no additional information compared to the total survey versus the total elections.

3.4 Effect of demographical attributes on political preference

In the original survey `data`, `sex`, `age`, and `education` are significantly associated with political preferences.
The anonymization process, while determining the loss of significance for all 3 attributes (at the 5% level), still ensures a set of fairly low raw p-values (0.020 for `age_group`, 0.070 for `sex`, and 0.052 for `education`), which all settle around 0.100 when adjusted.

Women result more likely than men to vote Red (42.39 % compared to 27.78%) and younger respondents result more heavily in favor of Green than older age bands.
Students from the three main compartments of University education (Vocational) Bachelors, Masters, and PhD are the main Red party strongholds, with a support between 50% and 57%.

3.5 Effect of demographical attributes on choice of voting channel

A statistically significant correlation with the choice of the voting channel emerged in the original survey for both `marital_status` and `age`. After data processing, the relationship with the latter is retained even after the shift to `age_group` (p-value of 0.027), while the latter barely falls above the threshold (p-value of 0.053).
From the side of utility, these are very acceptable results.

Only in the youngest age band (18-34) we register a preference of electronic voting means over paper (61% to 39%, respectively). Shifting on `marital_status`, only people who never married voted more electronically than via paper ballots, although barely (52% to 48%, respectively).

3.6 Reflections on the Utility-Risk trade-off

It is important to reiterate how we do observe some cases of feature relationships fading due to the anonymization process.
Therefore, while the processed survey data does retain some utility, since statistical significance is preserved for some pairings, and weakened relationships retain a reasonably low p-value, we can conclude that risk minimizing is prioritized throughout the described process.
We deem the amount of lost utility acceptable, but noticeable, and we renew our request for data analysts to consider using a slightly less strict threshold to detect statistical significance.